

Rigid Analytic Geometry Week 8 Solution

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Problem 1

Problem 2

Let $(x_n)_\mathbb{N}$ be a Cauchy sequence in K , then we have,

$$r_n = \sup_{m \geq n} |x_m - x_n|$$

is converging to 0. Let us define,

$$\mathfrak{L} = \{B_n = B(x_n, r_n) \mid n \in \mathbb{N}\}.$$

Then this is linearly ordered by inclusion. Thus by the assumption $x \in \bigcup_{\mathbb{N}} B_n$ exists. Thus $x_n \rightarrow x$ in K . K is complete.

Problem 3

Let $\mathfrak{L}$ be a linearly ordered family of balls. Since the valuation is discrete, we assume that it takes values in $\mathbb{Z}_{\geq 0}$. Thus we can take a minimum radius in $\mathfrak{L}$. By the assumption, the ball with the minimum radius is contained in the all the other balls.

Problem 4

Let $D = \{d_1, \dots, d_n, \dots\}$ be a countable dense subset of K . Let $B_1 = B(0, |d_1|)$. Then $d_1 \notin B_1$. We then inductively define B_n by such that B_{n+1} has a center in B_n and with a smaller radius so it does not contain d_{n+1} . Since $|K^\times|$ is dense, this construction is indeed possible.

Let $x \in \bigcup_n B_n$, then by the openness of each B_n and that we can take d_m which is arbitrary close to x , $d_m \in B_m$ for some m . This is a contradiction, thus the intersection is empty.

0.1 Tutor's solution

Problem 1 Check of definition.

Problem 5 Let $\mathcal{M}$ be a mock circle. By remark $f = gh$ where $g \in \mathbb{T}_1^\times$ and h is a polynomial. In particular, f has finitely many zeros $\alpha_1, \dots, \alpha_n$. Since,

$$\bigcup_{B \in \mathcal{M}} B = \emptyset,$$

there is $B \subseteq \mathcal{M}$ such that $\alpha_i \notin B$ for all $1 \leq i \leq n$. By Sheet 4 Problem 3, $|f(x)|$ is constant on B .

Problem 6 If $\|f_0\|_m \geq \|f_i\|_m$ for all $1 \leq i \leq n$, by the definition of $\Omega = \mathcal{R}(f_0|f_1, \dots, f_n)$, pick $B \in \mathcal{M}$ such that f_i are constant on B . Then $B \subseteq \Omega$.

If $\|f_i\|_{\mathcal{M}} > \|f_0\|_{\mathcal{M}}$ for some $1 \leq i \leq n$, take $B \in \mathcal{M}$ such that f_i is constant on B . By definition $B \cap \Omega = \emptyset$. it remains to show that $\xi_{\mathcal{M}}$ is a van der Put Point. Note that $K^\circ \in \xi_{\mathcal{M}}$. If $\Omega \subseteq \Theta$ are rational open such that $\Omega \in \xi_{\mathcal{M}}$, then there is $B \in \mathcal{M}$ such that $B \subseteq \Omega$. Thus $B \subseteq \Theta$. If $\Omega, \Theta \in \xi_{\mathcal{M}}$ then there is $B_1 \subseteq \Omega, B_2 \subseteq \Theta$. Therefore $\min(B_1, B_2) \subseteq \Omega \cap \Theta$. Thus $\Omega \cap \Theta \in \xi_{\mathcal{M}}$. if $\Omega = \Theta_1 \cup \dots \cup \Theta_n$, then $\Omega \in \xi_{\mathcal{M}}$, need to prove that $\Theta_i \in \xi_{\mathcal{M}}$ for some i . If this is not the case then for each $1 \leq i \leq n$, there are $B_i \subseteq \mathcal{M}$ and $B_i \cap \Theta_i = \emptyset$. Therefore,

$$\min(B_i) \cap (\Theta_1 \cup \dots \cup \Theta_n) = \emptyset.$$

This is a contradictino.

Problem 8 Let j be the largest index with $|f_j|r^j = \max_i |f_i|r^i$. Take $R_0 \in (r, 1)$. For $l > j$, since,

$$|f_i|r^i \rightarrow 0.$$

So there are only finitely many relevant terms $l \leq N$. Thus we can choose $R > r$ close to r enough such that

$$|f_l||x|^l < |f_j||x|^j$$

for $l \leq N, r < |x| < R$. For $l < j$, $|f_l|r^l \leq |f_j|r^j \leq n$. Therefore,

$$|f_l||x|^l \leq n \left(\frac{|x|}{r} \right)^l < n \left(\frac{|x|}{r} \right)^j.$$

Problem 9

Let A be a nat-ring and $X \subseteq A$ is a topologically nilpotent subset. The ideal generated by X is

$$X' = \left\{ \sum n_\alpha x_\alpha \mid x_\alpha \in X \right\}.$$

For each open ideal I , there is $N \in \mathbb{N}$ such that $\forall n > N, X^n \subseteq I$. Note $(X')^n$ is a finite combination of $m_{\underline{\alpha}} x_{\underline{\alpha}}$ where $\underline{\alpha}$ is an index. Also $x_1, \dots, x_l \in I$ so finite combination of them are also in I , hence $(X')^n \subseteq I$. By definition, X' is topologically nilpotent.